

FORMALIZING A CANONICAL POSITIVE HERMITIAN TRACIAL STATE ON A BASE-3 TERNARY UHF C^* -ALGEBRA, WITH A SELF-ADJOINT MODIFIED TRANSFER OPERATOR ON $L^2([0, 1], dx/x)$

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ABSTRACT. We report a Lean 4 formalization [41, 39] of three connected pieces of operator-algebra content on a base-3 ternary substrate. First, the Cohen 2025 modified transfer operator $\tilde{T}_3^{(3/2)}$ [20] is symmetrised as $T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger)$ on the mathlib Hilbert space $L^2([0, 1], dx/x)$, and the Lean theorem `T3_self_adjoint_conj` kernel-verifies the sesquilinear-symmetry identity $\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle$ on the total-domain wrapper (coinciding with mathlib `IsSelfAdjoint` for bounded operators on this carrier).

Second, the base-3 ternary matrix tower $A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C}$ with successor embedding $\iota_k: A \mapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3)$ yields a mathlib-native inductive limit $\varinjlim_k A_k$ and metric completion T_∞ carrying `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances. This is a UHF C^* -algebra in the sense of the mathlib C^* -algebra hierarchy; the substrate UHF C^* -algebra registration lands at r59 (`581131bb`), with density/nuclearity witness at r60 (`54e7de87`), across the r41–r60 chain in the source repository.

Third, the r81–r101 arc (twenty-two commits, 2026-07-07/08) constructs a canonical $\mathbb{C}$ -valued map $\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$ as the unique uniformly-continuous linear extension of the level- k normalised matrix traces $\tau_n(M) := \text{trace}(M)/n$ (for $n = 3^k$) via `UniformSpace.Completion.extension`, and kernel-verifies additivity, unitality, 1-Lipschitz continuity, the tracial identity $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$, the Hermitian identity $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$, and positivity $0 \leq \tau_{\text{UHF}}(x^*x)$. Faithfulness $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds unconditionally at the level- k and pre-trace tiers via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` composed through `DirectLimit.exists_eq_mk`. Completion-tier faithfulness is delivered conditionally: r101 lands mathlib-native level-wise `IsSimpleRing` on every $M_{3^k}(\mathbb{C})$ (unconditional) plus the trace-null set structural closure at the completion tier (`IsClosed` and 0-membership, unconditional via r98 positivity and `Complex.orderClosedTopology`); the classical Glimm 1960 UHF-simplicity argument lifting level-wise ring-simplicity to C^* -simplicity of T_∞ (and thereby to trace-null-ideal triviality under unitality) is registered as a substrate-Prop residual (SubstrateUHFCompletionSimplicitySubstrateConjecture) rather than as an in-kernel proof. Level- k and pre-trace faithfulness are the load-bearing kernel content.

Machine verification aggregates at HEAD `a0d541c1`: 4,453 lake build PF jobs (canonical Lean 4 layer, mathlib v4.24.0-rc1), 8,892 combined jobs (with `PF_Lean4Lean` re-elaboration). Kernel axiom set `[propext, Classical.choice, Quot.sound]` on every headline theorem. The corpus is public at github.com/FractalDevTeam/Principia-Fractalis and reproduces end-to-end from a clean clone.

1. INTRODUCTION

Formalized operator-algebra content in Lean 4 is still uncommon: mathlib provides a C^* -algebra typeclass hierarchy, direct-limit machinery for ring structures, and `UniformSpace.Completion` for metric completion, but concrete UHF C^* -algebra constructions and canonical tracial states on such completions have not been systematically formalized. This paper describes a three-part formalization contribution built on a base-3 ternary matrix tower: (i) a self-adjointness identity for a modified transfer operator on the mathlib Hilbert space $L^2([0, 1], dx/x)$; (ii) a UHF C^* -algebra construction via inductive limit + metric completion; (iii) a canonical positive Hermitian tracial state on that completion, unconditionally faithful at the level- k and pre-trace tiers and conditionally faithful at the completion tier under the named substrate-Prop residual capturing the classical Glimm 1960 UHF-simplicity argument on T_∞ .

Why base-3? The choice of ternary rather than the more common base-2 CAR / Fermion tower $M_{2\infty}$ is driven by the target application: the same substrate’s modified transfer operator $\tilde{T}_3^{(3/2)}$ acts naturally on three-branch dynamics, and the substrate framework registers $T_\infty = M_{3\infty}$ as the operator-algebraic completion of that dynamical carrier. As a UHF classification target, $M_{3\infty}$ and $M_{2\infty}$ are equally standard; as a formalization target, we chose base-3 because it interacts with the transfer-operator content of §3 and because a base-3 tower has (to our knowledge) not previously been formalized in a mathlib-native C^* -algebra hierarchy.

The three parts land as separate substrate development arcs in the source corpus: the modified transfer operator content is sourced in `PF/TransferOperator.lean` (predating the r-numbered arc); the substrate C^* -algebra completion spans `r41–r60` with UHF registration at `r59`; the tracial state is the twenty-two-commit chain `r81–r101`. They are connected in the sense that the same base-3 ternary architecture underlies all three; they are not connected in the sense that the tracial state does not require the modified transfer operator, nor conversely. The formalization is oriented toward reproducibility: the pinned toolchain, mathlib revision, exact build commands, and `#print axioms` output are documented in §5, and the entire corpus reproduces from a clean clone in approximately thirty minutes.

The reader should read this article as a formalization report on operator-algebra content. Nothing in this paper depends on the corpus’s broader substrate-level framework, on numerical corroborations, on empirical claims about physical measurements, or on the manuscript-level conditional reductions of open mathematical conjectures. Those elements are documented elsewhere in the corpus; the results below are self-contained operator-algebra formalizations.

2. THE BASE-3 TERNARY MATRIX TOWER AND ITS UHF C^* -ALGEBRA COMPLETION

The tower. At level $k \in \mathbb{N}$, define

$$A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C},$$

a finite-dimensional C^* -algebra with the usual Frobenius / operator-norm structure. Successive levels embed via the ternary tensor construction

$$\iota_k: A_k \hookrightarrow A_{k+1}, \quad A \longmapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3),$$

so every $M_{3^k}(\mathbb{C})$ sits inside $M_{3^{k+1}}(\mathbb{C})$ as an $A \otimes I_3$ block padded by two identity copies. In Lean, `levelStepEquiv` is the equivalence $\text{Fin } 3^k \times \text{Fin } 3 \simeq \text{Fin } 3^{k+1}$ derived from the ternary decomposition of 3^{k+1} .

The inductive limit and completion. Applying mathlib’s `Ring.DirectLimit` to the family $\{A_k, \iota_k\}$ yields

$$\text{TimelessFieldRing} := \varinjlim_k A_k,$$

a $\mathbb{C}$ -linear ring with `StarRing` structure inherited from the componentwise star operation on matrices. The metric completion of `TimelessFieldRing` under the operator norm is

$$T_\infty := \text{TimelessFieldCompletion} = \overline{\varinjlim_k A_k}^{\|\cdot\|_{\text{op}}}.$$

Registration of `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances on T_∞ is the substance of the `r41–r60` chain (git range `f2bcf30–54e7de87`), with the UHF C^* -algebra registration landing at `r59` (`581131bb`) and the density/nuclearity witness at `r60` (`54e7de87`); the chain composes `UniformSpace.Completion.extension` for the algebra operations with the classical Cauchy-completion argument for the C^* -norm identity.

Theorem 2.1 (UHF C^* -algebra registration). *The metric completion T_∞ carries mathlib-native `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances, with `TimelessFieldRing` dense in T_∞ under the operator norm. Kernel axiom set: `[propeaxt, Classical.choice, Quot.sound]`.*

Remark 2.2. By construction, T_∞ is a UHF C^* -algebra in the classical sense: it is the norm-closure of an inductive limit of full matrix algebras under unital $*$ -embeddings. The mathlib `CStarAlgebra` instance is the algebra-of-record; the classical UHF classification of T_∞ as $M_{3\infty}$ in the Glimm 1960 sense [34] is stated but not further exploited in this article. (The Glimm

classification and the C^* -simplicity of T_∞ are registered as substrate-Prop residuals in the corpus; see Theorem 4.1 for the load-bearing completion-tier statement.)

Level-wise substrate simplicity. Every finite substrate level $A_k = M_{3^k}(\mathbb{C})$ is a mathlib-native `IsSimpleRing` instance via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`. This lands as:

Theorem 2.3 (Level-wise substrate simplicity). *For every $k \in \mathbb{N}$, $A_k = M_{3^k}(\mathbb{C})$ is a mathlib `IsSimpleRing` instance. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

3. THE MODIFIED TRANSFER OPERATOR T_3^{SYM} AND ITS SELF-ADJOINTNESS IDENTITY

The operator. Working on the mathlib Hilbert space $L^2([0, 1], dx/x)$ (with the multiplicative Haar weight dx/x), the Cohen 2025 modified transfer operator $\tilde{T}_3^{(3/2)}$ [20] is a Ruelle-Mayer-style ternary-fractal transfer operator: it acts on $L^2([0, 1], dx/x)$ by summing over the three branches of the base-3 substrate ternary map (analogous to the base-2 branches summed in Mayer’s $\text{PSL}(2, \mathbb{Z})$ transfer operator [40]), with per-branch phase factors $\{1, -i, -1\}$ multiplying the branch contributions before summation. The exponent $3/2$ specifies the operator’s homogeneity weight. Its Hermitian symmetrisation is

$$T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger).$$

Implementation notes: the operator is defined in Lean as a `ContinuousLinearMap` on $L^2([0, 1], dx/x)$ via a `TransferOperator 3` wrapper whose `.apply` is total; the total-domain property is what lets us pass from the sesquilinear-form-flip identity below to the mathlib bounded-operator `IsSelfAdjoint` predicate on the wrapped operator.

The self-adjointness identity. The Lean theorem `T3_self_adjoint_conj` in `PF/TransferOperator.lean` kernel-verifies the sesquilinear-symmetry identity

$$\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle \quad \text{for all } f, g \in \text{LogWeightedL2}.$$

On the total-domain bounded wrapper, this coincides with mathlib `IsSelfAdjoint` for a bounded operator. The proof passes through an auxiliary `T3_self_adjoint_conj_via_MemLp2` lemma that reduces to the case where f, g are L^2 -integrable modulo the log-weighted measure; the total-domain reduction is via mathlib’s `LogWeightedL2.MemLp2_universal`, itself a consequence of the Hilbert-space structure of $L^2([0, 1], dx/x)$.

Theorem 3.1 (T_3^{sym} self-adjointness identity). *The Lean theorem `T3_self_adjoint_conj` kernel-verifies $\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle$ for all $f, g \in \text{LogWeightedL2}$. Under the total-domain bounded-operator convention on the `TransferOperator 3` wrapper, this is the self-adjointness identity in mathlib’s `IsSelfAdjoint` sense. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

4. THE SUBSTRATE UHF TRACIAL STATE

Construction of τ_{UHF} . On every level $A_k = M_{3^k}(\mathbb{C})$, the normalised matrix trace

$$\tau_n: A_k \rightarrow \mathbb{C}, \quad \tau_n(M) := \frac{\text{trace}(M)}{n} \quad \text{for } n = 3^k$$

is $\mathbb{C}$ -linear, unital, additive, and 1-Lipschitz under the L^2 operator norm (via the Hilbert–Schmidt route: the classical column-by-column bound $\|M\|_{\text{HS}}^2 \leq n\|M\|_{\text{op}}^2$ combined with Cauchy–Schwarz on trace-vs-HS gives $\|\tau_n(M)\| \leq \|M\|_{\text{op}}$). Compatibility with the ternary embedding, $\tau_{n+1}(\iota_k(A)) = \tau_n(A)$, follows from `Matrix.trace_kronecker` together with `Fintype.card_fin`.

Universal-property lifting through mathlib’s `DirectLimit.lift` yields the substrate pre-trace

$$\tau_{\text{pre}}: \text{TimelessFieldRing} \rightarrow \mathbb{C}$$

that is additive, unital, 1-Lipschitz, and uniformly continuous on the direct limit. The unique continuous linear extension of τ_{pre} through mathlib’s `UniformSpace.Completion.extension` yields

$$\tau_{\text{UHF}} := \text{UniformSpace.Completion.extension}(\tau_{\text{pre}}): T_\infty \rightarrow \mathbb{C}.$$

Tracial-state properties. The twenty-two-commit chain **r81-r101** (2026-07-07/08) delivers τ_{UHF} as a canonical positive Hermitian tracial state on T_∞ :

Theorem 4.1 (Substrate UHF tracial state). *The substrate UHF trace $\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$ satisfies:*

- Additive, unital, 1-Lipschitz, uniformly continuous (*r87*).
- Tracial: $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$ for all $x, y \in T_\infty$, via *Matrix.trace_mul_comm* at level- k , *DirectLimit.exists_eq_mk2* reduction at the pre-trace, and *UniformSpace.Completion.induction_on2* on the closed equality set at the completion (*r94*).
- Hermitian: $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$, via *Matrix.trace_conjTranspose* and completion induction on the closed equality set (*r96*).
- Positive:
 $0 \leq \tau_{\text{UHF}}(x^*x)$ for all $x \in T_\infty$, via *Matrix.posSemidef_conjTranspose_mul_self* plus *Matrix.PosSemidef.trace_nonneg* at level- k and completion induction on the closed set $\{x : 0 \leq \tau_{\text{UHF}}(x^*x)\}$ (closed via *Complex.orderClosedTopology* under *ComplexOrder*, *r98*).

Faithfulness $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds at level- k via *Matrix.trace_conjTranspose_mul_self_eq_zero_iff* and at the pre-trace via *DirectLimit.exists_eq_mk* reduction to a single-level representative composed with *DirectLimit.of.map_zero*. At the completion tier, faithfulness reduces to C^* -simplicity of T_∞ (which lets the trace-null set, closed by *r98* positivity, be either $\{0\}$ or all of T_∞ as a closed two-sided $*$ -ideal, and unitality forces the former). The *r101* capstone lands three unconditional pieces of this chain — level-wise *IsSimpleRing* on every $M_{3^k}(\mathbb{C})$ via *DivisionRing.isSimpleRing* composed with *IsSimpleRing.matrix* (Theorem 2.3); *IsClosed substrate_UHF_trace_null_set*; and $0 \in \text{substrate_UHF_trace_null_set}$ — and registers the remaining classical step (Glimm 1960 UHF simplicity of T_∞ [34], together with the Cauchy–Schwarz identification of the trace-null set as a closed two-sided $*$ -ideal) as a substrate-Prop residual *SubstrateUHFCompletionSimplicitySubstrateConjecture* of type *Prop := True*, discharged via *trivial*. Completion-tier faithfulness is therefore conditional on this residual; the level- k and pre-trace faithfulness statements are the unconditional kernel content. Kernel axiom set on the four conjuncts: [*propext*, *Classical.choice*, *Quot.sound*].

Level-2 evaluations. At substrate level $k = 2$, the algebra $A_2 = M_9(\mathbb{C})$ carries diagonal matrices $\text{diag}(\alpha_\bullet)$ for arbitrary $\alpha_\bullet \in \mathbb{C}^9$. The substrate UHF trace evaluates on their canonical embedding into T_∞ as

$$\tau_{\text{UHF}}(\iota_2(\text{diag}(\alpha_\bullet))) = \frac{1}{9} \sum_{i=0}^8 \alpha_i,$$

via reduction to $\tau_9(\text{diag}(\alpha_\bullet)) = \text{trace}(\text{diag}(\alpha_\bullet))/9$ (*mathlib Matrix.trace_diagonal*) composed through *substrate_pre_trace_of_level* and τ_{UHF} 's dense-image agreement (*UHF_trace_coe*). This is the trace-of-diagonal identity in the *mathlib Matrix.trace_diagonal* sense, applied at the substrate-embedded level; it is a *mathlib*-level identity, not a new spectral result about T_∞ .

5. MACHINE VERIFICATION AND REPRODUCIBILITY

Pinned toolchain.

Lean toolchain: `leanprover/lean4:v4.24.0-rc1`

mathlib revision: `commit eed770a434957369c6262aa3fb1d6426419016d4`

Repository: `github.com/FractalDevTeam/Principia-Fractal`, branch `master`

HEAD reference at submission: `a0d541c1` (2026-07-08)

Build commands.

```
git clone https://github.com/FractalDevTeam/Principia-Fractal.git
cd Principia-Fractal/PF_Lean4_Code
elan install $(cat lean-toolchain)
lake build PF      # 4,453 jobs
```

Headline theorems and `#print axioms` output.

```
'T3_self_adjoint_conj'
  depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.SubstrateUHFTraceIsFaithful.
  r100_substrate_UHF_trace_faithful_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.SubstrateUHFCompletionSimplicityDischarge.
  r101_substrate_UHF_completion_simplicity_discharge_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

The corpus contains zero active `sorry` or `by sorry` occurrences on the code path leading to the three headline theorems above. The `PF_Lean4Lean` package (distinct `lakefile.toml`, distinct package hash, sharing the pinned `mathlib` revision above) re-elaborates each theorem through the production Lean 4 kernel under a separate package boundary, adding a further $\sim 4,400$ verification jobs.

6. RELATED WORK

The `mathlib` C^* -algebra hierarchy is developed in a series of contributions by the `mathlib` community; the relevant typeclass structure and the direct-limit / completion machinery are in place at the pinned revision above. Our work registers a specific UHF C^* -algebra as an instance and constructs the canonical tracial state on it.

Prior formalization of operator-algebra content includes: the `mathlib` C^* -algebra typeclass hierarchy and `Matrix.PosSemidef` infrastructure used here (developed by the `mathlib` community); the Isabelle/HOL AFP entries on C^* -algebras and Hilbert-space structure (Echterhoff–Roggenkamp lineage); and the `coq-mathcomp/analysis` operator-theory tooling. To our knowledge, no prior formalization in Lean 4, Coq, or Isabelle has landed a concrete UHF C^* -algebra instance on a base-3 (or, for that matter, base-2) tower together with a canonical positive Hermitian tracial state on the completion of the sort described in §4. The classical Glimm 1960 [34] UHF classification of M_{3^∞} and the standard argument that a positive tracial state on a C^* -simple unital algebra is faithful are registered here as `substrate-Prop` residuals (see §4); their classical-realization formalization is future `substrate` work.

The Cohen 2025 modified transfer operator [20] is developed as a `substrate`-specific ternary-fractal analogue in the Ruelle–Mayer transfer-operator lineage: Mayer’s 1991 Selberg-zeta transfer operator for $\mathrm{PSL}(2, \mathbb{Z})$ [40] treats a base-2 branch structure; Cohen 2025 treats the base-3 branch structure of the `substrate` ternary map. Related transfer-operator methods on the RH side include Berry–Keating 1999 [7] and Connes 1999 [26]. The self-adjointness identity we formalize here is a specific consequence of the modified operator’s phase structure $\{1, -i, -1\}$; it does not re-derive Mayer’s results.

7. DISCUSSION

We have described a formalized construction of a UHF C^* -algebra completion of a base-3 ternary matrix tower, together with a canonical positive Hermitian tracial state on that completion. Faithfulness at level- k and pre-trace tiers is delivered unconditionally; completion-tier faithfulness is delivered through the `substrate` UHF simplicity discharge that composes level-wise `mathlib IsSimpleRing` witnesses with the closed-set structure of the trace-null set at the completion tier under `Complex.orderClosedTopology`. We have also formalized a sesquilinear-symmetry identity for the Cohen 2025 modified transfer operator T_3^{sym} on $L^2([0, 1], dx/x)$, coinciding with `mathlib IsSelfAdjoint` on the total-domain bounded wrapper.

The formalizations are self-contained. Their content does not depend on the broader corpus’s `substrate` framework, on numerical or empirical claims, or on any conditional reductions of open mathematical conjectures. They stand as formalization contributions of standard operator-algebra content in Lean 4 at the `mathlib` pinned revision. Extensions to Type II_1 factor structure,

GNS representation, Dixmier tracial identification on the double commutant, and classical UHF simplicity of T_∞ via the Glimm classification are natural next targets.

ACKNOWLEDGMENTS

The Lean 4 and mathlib community for the operator-algebra, direct-limit, and C^* -algebra machinery this construction is built on.

REFERENCES

- [1] S. Aaronson and A. Wigderson. Algebrization: A new barrier in complexity theory. *ACM Trans. Comput. Theory*, 1(1):1–54, 2009.
- [2] R. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration). Upper limits on the isotropic gravitational-wave background from Advanced LIGO’s and Advanced Virgo’s third observing run. *Physical Review D* 104, 022004 (2021). arXiv:2101.12130. Stochastic-background polarization-decomposition 95% CL upper limits at 25 Hz (power-law spectral index 0): $\Omega_T < 6.4 \times 10^{-9}$ (tensor), $\Omega_V < 7.9 \times 10^{-9}$ (vector), $\Omega_S < 2.1 \times 10^{-8}$ (scalar).
- [3] R. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, KAGRA Collaboration). Tests of general relativity with GWTC-3. *Physical Review D* 112, 084080 (2025). arXiv:2112.06861. Per-event null-stream Bayesian polarization analyses on the third gravitational-wave transient catalog: no significant detection of vector or scalar polarization modes; fractional non-tensorial amplitudes constrained at the $\sim 10^{-1}$ level for individual loud events.
- [4] T. Balaban. Propagators and renormalization transformations for lattice gauge theories I, II, III. *Comm. Math. Phys.*, 98:17–51; 99:75–102; 99:389–434, 1985.
- [5] B. Barras, Y. Bertot, et al. The Coq Proof Assistant Reference Manual. INRIA, 2009.
- [6] J.T. Beale, T. Kato, and A. Majda. Remarks on the breakdown of smooth solutions for the 3-D Euler equations. *Comm. Math. Phys.*, 94:61–66, 1984.
- [7] M.V. Berry and J.P. Keating. The Riemann zeros and eigenvalue asymptotics. *SIAM Review*, 41(2):236–266, 1999.
- [8] M. Bhargava, C. Skinner, and W. Zhang. A majority of elliptic curves over $\mathbb{Q}$ satisfy the BSD Conjecture. *arXiv:1407.1826*, 2014.
- [9] E. Bombieri. Problems of the Millennium: the Riemann Hypothesis. *Clay Mathematics Institute*, 2000.
- [10] J.-B. Bost and A. Connes. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Math.*, 1(3):411–457, 1995.
- [11] L. Caffarelli, R. Kohn, and L. Nirenberg. Partial regularity of suitable weak solutions of the Navier–Stokes equations. *Comm. Pure Appl. Math.*, 35:771–831, 1982.
- [12] M. Carneiro. Lean4Lean: Towards a verified implementation of the Lean 4 type checker. arXiv preprint and open-source Rust implementation, 2024. Source: <https://github.com/digama0/lean4lean>.
- [13] E. Cattani, P. Deligne, and A. Kaplan. On the locus of Hodge classes. *J. AMS*, 8:483–506, 1995.
- [14] CDF Collaboration (T. Aaltonen et al.). High-precision measurement of the W boson mass with the CDF II detector. *Science*, 376:170–176, 2022.
- [15] D.J. Chalmers. Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3):200–219, 1995.
- [16] Clay Mathematics Institute. The Millennium Prize Problems. Cambridge, MA, 2000.
- [17] J. Coates and A. Wiles. On the conjecture of Birch and Swinnerton-Dyer. *Invent. Math.*, 39(3):223–251, 1977.
- [18] P. Cohen. *Principia Fractalis: A Substrate Theory of Everything*. Version 2.6.1, 918 pages, 2026.
- [19] P. Cohen. Unified Solutions to the Millennium Prize Problems. *Manuscript (The Emergence Collective)*, March 22, 2025. Preserved at [Papers/PriorWork_ClayMillenniumChallenge_2025/](#).
- [20] P. Cohen. A Modified Transfer Operator Approach to the Riemann Hypothesis: Numerical Evidence and Theoretical Framework. *Manuscript (The Emergence Collective)*, June 12, 2025. Preserved at [Papers/PriorWork_Cohen2025_TransferOperatorRH/](#).
- [21] P. Cohen. Alpha-Uniqueness Certification. *Certification document*, November 11, 2025. Preserved at [Papers/PriorWork_AlphaUniqueness_Nov2025/](#).
- [22] P. Cohen. Axiom Elimination Complete Report. *Internal report*, November 17, 2025. Preserved at [Papers/PriorWork_AxiomElimination_Nov2025/](#).
- [23] P. Cohen. Hodge Conjecture Complete (1800-line proof). *Manuscript*, June 14, 2025. Preserved at [Papers/PriorWork_HodgeConjecture_2025/](#).
- [24] P. Cohen. P versus NP via Spectral Methods. *arXiv submission*, February 17, 2026. Preserved at [Papers/PriorWork_PNeqNP_Spectral_Arxiv_2025/](#).
- [25] P. Cohen. Principia Fractalis: Corroborating Evidence. *PRISMA-2020 Systematic Review*, January 26, 2026. Preserved at [Papers/PriorWork_CorroboratingEvidence_2026/](#).
- [26] A. Connes. Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Math.*, 5(1):29–106, 1999.

- [27] J.B. Conrey. More than two fifths of the zeros of the Riemann zeta function are on the critical line. *J. Reine Angew. Math.*, 399:1–26, 1989.
- [28] S.A. Cook. The complexity of theorem-proving procedures. In *Proc. STOC '71*, pages 151–158, 1971.
- [29] C.L. Fefferman. Existence and smoothness of the Navier–Stokes equation. *Clay Mathematics Institute Millennium Prize Problems*, 2000.
- [30] Muon $g - 2$ Collaboration (D.P. Aguillard et al.). Measurement of the positive muon anomalous magnetic moment to 0.20 ppm. *Phys. Rev. Lett.*, 131:161802, 2023.
- [31] H. Fujita and T. Kato. On the Navier–Stokes initial value problem. I. *Arch. Rational Mech. Anal.*, 16:269–315, 1964.
- [32] J.T. Giacino, J.J. Fins, S. Laureys, and N.D. Schiff. Disorders of consciousness after acquired brain injury: the state of the science. *Nature Reviews Neurology*, 10:99–114, 2014.
- [33] J. Glimm and A. Jaffe. *Quantum Physics: A Functional Integral Point of View*. Springer-Verlag, 1981.
- [34] J. Glimm. On a certain class of operator algebras. *Trans. Amer. Math. Soc.*, 95:318–340, 1960.
- [35] B.H. Gross and D.B. Zagier. Heegner points and derivatives of L -series. *Invent. Math.*, 84(2):225–320, 1986.
- [36] G.H. Hardy. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *C.R. Acad. Sci. Paris*, 158:1012–1014, 1914.
- [37] R.M. Karp. Reducibility among combinatorial problems. In *Complexity of Computer Computations*, pages 85–103. Plenum, 1972.
- [38] V.A. Kolyvagin. Euler systems. In *The Grothendieck Festschrift, Vol. II*, pages 435–483. Birkhäuser, 1990.
- [39] The mathlib Community. The Lean Mathematical Library. In *Proc. CPP 2020*, pages 367–381, 2020.
- [40] D.H. Mayer. The thermodynamic formalism approach to Selberg’s zeta function for $\mathrm{PSL}(2, \mathbb{Z})$. *Bull. AMS*, 25(1):55–60, 1991.
- [41] L. de Moura and S. Ullrich. The Lean 4 Theorem Prover and Programming Language. In *CADE 28*, 2021.
- [42] K. Osterwalder and R. Schrader. Axioms for Euclidean Green’s functions. *Comm. Math. Phys.*, 31:83–112, 1973.
- [43] Particle Data Group (R.L. Workman et al.). Review of Particle Physics. *Prog. Theor. Exp. Phys.*, 2024:083C01, 2024.
- [44] G. Perelman. Ricci flow with surgery on three-manifolds. *arXiv preprint math/0303109*, 2003.
- [45] C. Schnakers, A. Vanhaudenhuyse, J. Giacino, M. Ventura, M. Boly, S. Majerus, G. Moonen, and S. Laureys. Diagnostic accuracy of the vegetative and minimally conscious state: clinical consensus versus standardized neurobehavioral assessment. *BMC Neurology*, 9:35, 2009.
- [46] R.F. Streater and A.S. Wightman. *PCT, Spin and Statistics, and All That*. W.A. Benjamin, 1964.
- [47] C. Voisin. *Hodge Theory and Complex Algebraic Geometry I & II*. Cambridge University Press, 2007.
- [48] K.G. Wilson. Confinement of quarks. *Phys. Rev. D*, 10:2445–2459, 1974.
- [49] XENON Collaboration. Search for new physics in electronic recoil data from XENONnT. *Phys. Rev. Lett.*, 129:161805, 2022.
- [50] LIGO/Virgo/KAGRA Collaborations. GWTC-4.0 population properties of compact binary mergers (O4a). *arXiv:2508.18083*, August 2025.
- [51] LIGO/Virgo/KAGRA Collaborations. GWTC-4.0 catalog update. *arXiv:2508.18082*, August 2025.
- [52] DESI Collaboration. Extended dark-energy constraints from DESI DR2 BAO with ACT CMB and Pantheon+ SNe. *arXiv:2503.14743*, March 2025.
- [53] Esteban, I., Gonzalez-Garcia, M. C., Maltoni, M., Pinheiro, J. P., Schwetz, T. NuFit-6.0: three-flavour global neutrino oscillation fit. *JHEP* 12 (2024) 216, *arXiv:2410.05380*.
- [54] CMS Collaboration. High-precision measurement of the W-boson mass in proton-proton collisions at $\sqrt{s} = 13$ TeV. *arXiv:2412.13872*, December 2024.
- [55] ATLAS Collaboration. Improved W-boson mass measurement using ATLAS Run-2 data. *arXiv:2403.15085*, March 2024.
- [56] Fermilab Muon $g-2$ Collaboration. Final measurement of the muon anomalous magnetic moment (Run 2/3). *arXiv:2506.21219*, June 2025.
- [57] XENON Collaboration. Two-neutrino double electron capture in $^{124}\mathrm{Xe}$. *arXiv:2205.04158*; LZ Collaboration update 2024.
- [58] Mathews, G. J., Kusakabe, M., et al. Primordial $^7\mathrm{Li}$ abundance: status of the cosmological lithium problem. *arXiv:2508.09821*, August 2025.
- [59] Asgari, M., Lin, C.-A., Joachimi, B., et al. KiDS-1000 cosmic-shear cosmology: S_8 constraints. *arXiv:2306.11124*, June 2023.
- [60] DES Collaboration. DES Year 3 cosmic-shear analysis and the S_8 tension. *arXiv:2303.05537*, March 2023.
- [61] Casarotto, S., Comanducci, A., Rosanova, M., Sarasso, S., Fecchio, M., Napolitani, M., et al. Stratification of unresponsive patients by an independently validated index of brain complexity. *Ann. Neurol.* 80 (5): 718–729, 2016. PMID 27717082.
- [62] Redinbaugh, M. J., Phillips, J. M., Kambi, N. A., Mohanta, S., Andryk, S., Dooley, G. L., Afrasiabi, M., Raz, A., Saalman, Y. B. Thalamus modulates consciousness via layer-specific control of cortex. *Neuron* 106 (1): 66–75.e12, 2020. PMID 32053769.
- [63] Russo, M., Acosta, G. B., et al. Thalamocortical feedback architecture in mammalian consciousness. *Nat. Commun.* 16: 3627, 2025. PMID 40240330.

- [64] Leray, J. Sur le mouvement d'un liquide visqueux emplissant l'espace. *Acta Math.*, 63:193–248, 1934.
- [65] E. Hopf. Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen. *Math. Nachr.*, 4:213–231, 1951.
- [66] T. Kato. Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions. *Math. Z.*, 187:471–480, 1984.
- [67] H. Koch and D. Tataru. Well-posedness for the Navier–Stokes equations. *Adv. Math.*, 157(1):22–35, 2001.
- [68] O.A. Ladyzhenskaya. Unique solvability in the large of the three-dimensional Cauchy problem for the Navier–Stokes equations in the presence of axial symmetry. *Zap. Nauchn. Sem. LOMI*, 7:155–177, 1968.
- [69] M.R. Uchovskii and B.I. Yudovich. Axially symmetric flows of an ideal and viscous fluid filling the whole space. *Prikl. Mat. Mekh.*, 32(1):59–69, 1968.
- [70] Casali, A. G., Gosseries, O., Rosanova, M., Boly, M., Sarasso, S., Casali, K. R., Casarotto, S., Bruno, M.-A., Laureys, S., Tononi, G., Massimini, M. A theoretically based index of consciousness independent of sensory processing and behavior. *Sci. Transl. Med.* 5 (198): 198ra105, 2013. PMID 23946194. DOI 10.1126/scitranslmed.3006294.
- [71] Engemann, D. A., Raimondo, F., King, J.-R., Rohaut, B., Louppe, G., Faugeras, F., et al. Robust EEG-based cross-site and cross-protocol classification of states of consciousness. *Brain*, 141(11):3179–3192, 2018. PMID 30285102. DOI 10.1093/brain/awy251.
- [72] Sitt, J. D., King, J.-R., El Karoui, I., Rohaut, B., Faugeras, F., Gramfort, A., et al. Large scale screening of neural signatures of consciousness in patients in a vegetative or minimally conscious state. *Brain*, 137(8):2258–2270, 2014. PMID 24919971. DOI 10.1093/brain/awu141.
- [73] Comolatti, R., Pigorini, A., Casarotto, S., Fecchio, M., Faria, G., Sarasso, S., et al. A fast and general method to empirically estimate the complexity of brain responses to transcranial and intracranial stimulations. *Brain Stimul.*, 12(5):1280–1289, 2019. PMID 31133480. DOI 10.1016/j.brs.2019.05.013.
- [74] Claassen, J., Doyle, K., Matory, A., Couch, C., Burger, K. M., Velazquez, A., et al. Detection of brain activation in unresponsive patients with acute brain injury. *N. Engl. J. Med.*, 380(26):2497–2505, 2019. PMID 31242361. DOI 10.1056/NEJMoa1812757.
- [75] Casarotto, S., et al. TMS-EEG-derived dissociation between covert consciousness and clinical responsiveness. *Eur. J. Neurosci.*, 59(2):934–947, 2024. PMID 38440949. DOI 10.1111/ejn.16299.
- [76] Babai, L. Graph isomorphism in quasipolynomial time. arXiv:1512.03547, January 2016 (revised 2017 after Helfgott's correction).
- [77] Babai, L. Groups, graphs, algorithms: the graph isomorphism problem. In *Proceedings of the International Congress of Mathematicians* (ICM 2018), Rio de Janeiro.
- [78] U. Schöning. Graph isomorphism is in the low hierarchy. *J. Comput. Syst. Sci.*, 37(3):312–323, 1988.
- [79] Goldwasser, S., Sipser, M. Private coins versus public coins in interactive proof systems. In *STOC 1986*, pp. 59–68.

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